



8633 - Back from the dead or the final nail in the coffin: is SN2008S still alive?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Conor Ransome (PI)	Harvard University
Prof. Ashley Villar (CoI)	Harvard University
Dr. Ori Dosovitz Fox (CoI)	Space Telescope Science Institute
Prof. Alex V. Filippenko (CoI)	University of California - Berkeley
Dr. Alexander Thomas Gagliano (CoI)	Massachusetts Institute of Technology
Dr. Sebastian Gomez (CoI)	University of Texas at Austin
Dr. Yize Dong (CoI)	Center for Astrophysics Harvard & Smithsonian
Dr. Anya Elizabeth Nugent (CoI)	Center for Astrophysics Harvard & Smithsonian
S. Karthik Yadavalli (CoI)	The Pennsylvania State University
Kaylee Marie de Soto (CoI)	The Pennsylvania State University

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	2008S NIRcam	NIRCam Imaging	(1) SN-2008S
	2	2008S MIRI	MIRI Imaging	(1) SN-2008S

ABSTRACT

SN2008S is the prototype for a class of underluminous, dusty transients whose true origins are unclear. 08S-like transients are often regarded as supernova impostors, non-terminal mass loss eruptions of massive stars. However this may not be the case, pre-explosion imaging of SN2008S shows that it may have a low mass progenitor and Spitzer follow-up found that SN2008S had faded below the luminosity of its progenitor, perhaps indicating a terminal explosion. However, a surviving progenitor may exist and be dust-enshrouded. This proposal aims to finally solve the mystery

JWST Proposal 8633 (Created: Monday, September 15, 2025, 2:00:15PM Eastern Standard Time) - Overview

of this common class of underluminous (if true supernovae) transient. Using NIRcam and MIRI over a range of filters, we aim to see if the transient is still detectable, construct spectral energy distributions in order to fit models of dust enshrouded progenitors in order to determine if dust could feasibly still be obscuring a surviving progenitor. Furthermore, SEDs of the stars local to SN2008S can be used to estimate the mass, informing us on the mass of the progenitor of SN2008S. The timing of this proposal is perfect as we are on the cusp of an explosion in transient discoveries with LSST at the Vera C. Rubin Observatory coming online in the next year or so. LSST will discover enormous numbers of transients and this investigation will provide a vital case-study for these mysterious objects.

OBSERVING DESCRIPTION

We aim to use NIRcam and MIRI in the following filters: F460M, F560W, F1000W, F1280W, F1500W and F2100W in order to determine whether SN2008S has faded so much that feasible dust behaviour can no longer hide a surviving progenitor or if there is cold dust obscuring the progenitor. Hence this will determine whether SN2008S is a true supernova or an impostor.

Proposal 8633 - Targets - Back from the dead or the final nail in the coffin: is SN2008S still alive?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SN-2008S	RA: 20 34 45.3300 (308.6888750d) Dec: +60 05 58.40 (60.09956d) Equinox: J2000	Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Eruptive variables, Supernovae] Extended=NO				

Proposal 8633 - Observation 1 - Back from the dead or the final nail in the coffin: is SN2008S still alive?

Observation	Proposal 8633, Observation 1: 2008S NIRcam										Mon Sep 15 19:00:15 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SN-2008S	RA: 20 34 45.3300 (308.6888750d)			Epoch of Position: 2000					
			Dec: +60 05 58.40 (60.09956d)								
			Equinox: J2000								
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Eruptive variables, Supernovae] Extended=NO									
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULE		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F200W	F460M	DEEP2	5	1	4	4	3521.661		
Special Requirements	Offset 25.0 arcsec, 25.0 arcsec										

Proposal 8633 - Observation 2 - Back from the dead or the final nail in the coffin: is SN2008S still alive?

Observation	Proposal 8633, Observation 2: 2008S MIRI										Mon Sep 15 19:00:15 GMT 2025
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	SN-2008S	RA: 20 34 45.3300 (308.6888750d) Dec: +60 05 58.40 (60.09956d) Equinox: J2000			Epoch of Position: 2000					
Template	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	Category=Star Description=[Eruptive variables, Supernovae] Extended=NO										
Dithers	Subarray										
	FULL										
Spectral Elements	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		1	1			DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	20	4	1	Dither 1	4	16	921.313	
	2	F1000W	FASTR1	50	4	1	Dither 1	4	16	2253.332	
	3	F1500W	FASTR1	70	12	1	Dither 1	4	48	9446.236	